Exercise Sheet 1

Introduction to Linear Programming

Algorithms and Geometry of Linear Programming — DIMAG Summer School, August 2026
Lecturer: D. Dadush

This sheet restates the four exercises of Lecture 1; the results of the lecture may be used freely. Throughout, $\mathbf{A} \in \mathbb{R}^{m \times n}$, and $a_i^\mathsf{T} := \mathbf{A}_{i,\bullet}$ denotes its i-th row.

Exercise 1 (Convexity of hulls, cones and sums)**.** For finite sets $V, R \subseteq \mathbb{R}^n$, recall the *convex hull* and *generated cone*

$$\operatorname{conv}(V) := \Big\{ \sum_{v \in V} \lambda_v v \;:\; \lambda \geq \mathbf{0},\ \sum_{v \in V} \lambda_v = 1 \Big\}, \qquad \operatorname{cone}(R) := \Big\{ \sum_{r \in R} \mu_r r \;:\; \mu \geq \mathbf{0} \Big\},$$

and the *Minkowski sum* $A + B := \{a + b : a \in A,\ b \in B\}$ of sets $A, B \subseteq \mathbb{R}^n$. Recall also that a set $C \subseteq \mathbb{R}^n$ is a *cone* if it is convex and closed under nonnegative scaling: $\lambda x \in C$ for all $x \in C$, $\lambda \geq 0$. Prove the following convexity properties:

(a) $\operatorname{conv}(V)$ is convex and $\operatorname{cone}(R)$ is a cone;

(b) $A + B$ is convex whenever $A, B \subseteq \mathbb{R}^n$ are convex.

Exercise 2 (Carathéodory's theorem, conic version)**.** In this exercise, subscripts on $\mathbf{A}$ select *columns*: $\mathbf{A}_S := \mathbf{A}_{\bullet,S}$, the column submatrix indexed by S, and $\mathbf{A}_j := \mathbf{A}_{\bullet,j}$ is the j-th column. The *support* of a vector $x \in \mathbb{R}^k$ is the set of coordinates on which it is nonzero, $\operatorname{supp}(x) := \{j : x_j \neq 0\}$. Let $\mathbf{A} \in \mathbb{R}^{n \times k}$ and let $y = \mathbf{A}x$ for some $x \geq \mathbf{0}$, so that y is a nonnegative combination of the columns of $\mathbf{A}$ with coefficients x. Show that there exists $x' \geq \mathbf{0}$ with $y = \mathbf{A}x'$ whose support columns $\{\mathbf{A}_j : j \in \operatorname{supp}(x')\}$ are linearly independent; in particular, $|\operatorname{supp}(x')| \leq \operatorname{rank}(\mathbf{A})$. (Hint: Show that $Q := \{x \in \mathbb{R}^k : \mathbf{A}x = y,\ x \geq \mathbf{0}\}$ has a vertex, and that any vertex has the desired support bound.)

Exercise 3 (Complementary slackness)**.** Let x be primal feasible and y be dual feasible for the pair

$$\max\{c^\mathsf{T} x : \mathbf{A}x \leq b\} \qquad \text{and} \qquad \min\{b^\mathsf{T} y : \mathbf{A}^\mathsf{T} y = c,\ y \geq \mathbf{0}\}.$$

Show that x and y are both optimal if and only if $y_i(b_i - a_i^\mathsf{T} x) = 0$ for all $i \in [m]$, i.e., positive multipliers appear only on tight constraints. (Hint: Inspect where the proof of weak duality can be loose.)

Exercise 4 (Proof of the Farkas lemma)**.** Prove the Farkas lemma: the system $\mathbf{A}x \leq b$ is infeasible if and only if there exists $y \geq \mathbf{0}$ with $\mathbf{A}^\mathsf{T} y = \mathbf{0}$ and $b^\mathsf{T} y = -1$. (Hint: ($\Leftarrow$) Evaluate $y^\mathsf{T}\mathbf{A}x$ in two ways. ($\Rightarrow$) Apply strong duality to the "phase one" LP $\min\{\mathbf{1}^\mathsf{T} z : \mathbf{A}x - z \leq b,\ z \geq \mathbf{0}\}$, which measures the minimum total constraint violation, and normalize an optimal dual solution.)